

ABG Interpretation Workshop

Moderator Slide Deck - 13 Cases

13 cases x 15 min = approx 3.25 hours . Or select 8 cases for 2 hours.

One case, 15 minutes, two phases

Step 1 - All 13 cases handed out at once

Every participant receives the full case booklet at once.

Step 2 - 8-minute settling period

Participants may read ahead. Faculty do not teach yet.

Step 3 - Each case is one 15-minute unit

Small group - 8 min: 3 interpret, 3 discuss, 2 synthesise

Plenary - 7 min: 3 random draw, 3 expert comment, 1 take-home

The Unified 10-Step Method

Step	Tool / key question
1. Clinical history	What does the story alone suggest?
2. Internal validity	$[H^+] = 24 \times PaCO_2 / HCO_3^-$ - does it match pH? Arterial?
3. pH	Acidaemia, alkalaemia, or normal? A normal pH does not mean a normal gas.
4. Primary disorder	Does $PaCO_2$ or HCO_3^- explain the pH direction?
5. Expected compensation	Metabolic acidosis: $PaCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$. Metabolic alkalosis: $PaCO_2 = 0.7 \times HCO_3^- + 21 \pm 2$.
6. Anion gap	$Na^+ - (Cl^- + HCO_3^-)$; add 2.5 per 1 g/dL albumin below 4.0.
7. Delta ratio	$(AG - 12) / (24 - HCO_3^-)$ - only if AG raised.
8. Oxygenation & integration	P/F, OI, CaO_2 , $ScvO_2$, MetHb. Does the story fit the child?
9. SID / Stewart analysis	$SID = Na^+ - Cl^-$ (normal ~36). Low = chloride-driven acidosis. High = chloride-loss alkalosis. $Atot = \text{albumin} + \text{phosphate}$. Unmeasured anions = $AG - (\text{lactate} + \text{ketones})$.
10. Post-intervention re-evaluation	Has the mechanism changed? Recalculate AG, SID, $Atot$.

DKA

Small group - 8 min - Presenting gas

10-year-old girl, 2 weeks of polyuria and 3 days of vomiting and abdominal pain. Acidotic breathing present, haemodynamically stable, GCS 11.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.08	14	4	36	132
K	Cl	Albumin	Glucose	b-OHB
4.5	98	4.0	480	5.8

Plenary reveal - 7 min - Post-intervention gas

Intervention: 10 mL/kg NS bolus, insulin infusion, NS + KCl.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	26	11	34	142
K	Cl	Albumin	Glucose	b-OHB
3.4	119	4.0	230	0.6

DKA

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. **Primary disorder:** HCO_3^- 4 (low) with acidaemia -> metabolic acidosis.
5. **Expected compensation:** Winter's: $\text{PaCO}_2 = 1.5 \times 4 + 8 = 14 \pm 2$. Actual 14 - appropriate.
6. **Anion gap:** $\text{AG} = 132 - (98 + 4) = 30$ (markedly raised). Albumin normal - no correction needed.
7. **Delta ratio:** $(30 - 12)/(24 - 4) = 0.9$ -> pure HAGMA.
8. **Oxygenation & integration:** Severe HAGMA with maximal appropriate respiratory compensation; AG 30 fully explained by ketoacids (b-OHB 5.8). Severe DKA.

SID / Stewart

- SID:** $132 - 98 = 34$ - Low-normal - mild chloride-driven component already present.
- Atot:** Albumin 4.0 - Normal.
- Unmeasured anions:** $30 - 5.8 = 24.2$ - Markedly raised - ketoacids dominate.
- Mechanism:** - - Unmeasured anions, with mild chloride-driven component.

Post-intervention

2. **Internal validity:** $[\text{H}^+]$ ~56 nmol/L - internally valid.
3. **pH:** 7.25 - improving.
4. **Primary disorder:** HCO_3^- 11 -> metabolic acidosis.

5. Expected compensation: Winter's: $\text{PaCO}_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$. Actual 26 - appropriate.

6. Anion gap: $\text{AG} = 142 - (119 + 11) = 12$ (normal - HAGMA resolved).

7. Delta ratio: $(12 - 12)/(24 - 11) = 0 \rightarrow$ pure NAGMA.

9. SID / Stewart: $\text{SID} = 142 - 119 = 23$ (markedly low - chloride-driven). Unmeasured anions = $12 - 0.6 = 11.4$ (normal). Mechanism changed: ketoacids cleared, chloride load now driving.

DKA

Expert comment . answers

Q1. Calculate the corrected serum sodium.

Q2. What do the glucose and b-OHB values suggest?

A1. $\text{Na}^+ + 1.6 \times [(480 - 100)/100] \sim 138$. True hyponatraemia is dilutional from hyperglycaemia.

A2. Hyperglycaemic ketosis driving the gap - severe DKA.

Post-intervention case-specific answers

Q1. Compare this ABG with the initial ABG. What has changed?

Q2. Why is acidosis persisting despite improvement in glucose and b-OHB?

Q3. Calculate the corrected serum sodium.

A1. pH improved (7.08->7.25); PaCO₂ rose (14->26); HCO₃ rose modestly (4->11); glucose fell (480->230); b-OHB fell (5.8->0.6); Na rose (132->142); K fell (4.5->3.4); Cl rose sharply (98->119); AG normalised (30->12). The acid has changed character.

A2. Ketoacidosis resolved. Persisting low HCO₃ is now hyperchloraemic NAGMA from saline chloride load - expected, not treatment failure.

A3. $142 + 1.6 \times [(230 - 100)/100] \sim 144$.

Take-home

Recalculate the anion gap on every repeat gas - a persistently low HCO₃ after treatment can mean the acid has changed character.

Pyloric stenosis

Small group - 8 min - Presenting gas

6-week-old first-born boy with 10 days of progressively forceful, non-bilious vomiting after every feed. Sunken eyes, lethargic but arousable, still hungry after vomiting.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.52	48	38	34	132
K	Cl	Albumin	Glucose	Urine Cl
3.0	82	4.0	90	<15

Plenary reveal - 7 min - Post-intervention gas

Intervention: IV 0.9% saline + KCl, correct dehydration and hypokalaemia, then pyloromyotomy.

pH	PaCO ₂	HCO ₃	Na	K
7.45	40	27	138	4.0
Cl	Albumin	Urine Cl		
100	4.0	>20		

Pyloric stenosis

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. **Primary disorder:** HCO_3^- 38 (raised) with alkalaemia $\rightarrow$ metabolic alkalosis.
5. **Expected compensation:** $\text{PaCO}_2 = 0.7 \times 38 + 21 = 47.6 \pm 2$ (45.6-49.6). Actual 48 - appropriate.
6. **Anion gap:** $\text{AG} = 132 - (82 + 38) = 12$ (normal).
7. **Delta ratio:** Not applicable - no HAGMA.
8. **Oxygenation & integration:** Hypochloraemic, hypokalaemic metabolic alkalosis with appropriate respiratory compensation; low urine Cl confirms chloride-responsive process.

SID / Stewart

- SID:** $132 - 82 = 50$ - Markedly high - chloride-loss alkalosis.
- Atot:** Albumin 4.0 - Normal.
- Unmeasured anions:** $\text{AG} 12 - 0 = 12$ - Normal.
- Mechanism:** - - Loss of gastric HCl has removed chloride, raising SID.

Post-intervention

9. **SID / Stewart:** $\text{SID} = 138 - 100 = 38$ (normalising - chloride repletion working). Urine Cl now >20 confirms kidney no longer conserving chloride.

Pyloric stenosis

Expert comment . answers

Q1. What is the significance of the low urine chloride?

Q2. What is the likely clinical diagnosis?

A1. Urine Cl $< 15\text{-}20$ mmol/L confirms chloride-responsive metabolic alkalosis: kidney appropriately conserving chloride. Alkalosis will correct with chloride-rich fluid. Contrast with chloride-resistant causes (Bartter/Gitelman) where urine Cl is inappropriately high.

A2. Hypertrophic pyloric stenosis - persistent vomiting of gastric HCl and volume depletion produce hypochloraemic, hypokalaemic metabolic alkalosis; secondary hyperaldosteronism drives further renal H^+ and K^+ loss.

Take-home

A urine chloride confirms whether a metabolic alkalosis will respond to chloride repletion. SID tracks the response.

Gastroenteritis

Small group - 8 min - Presenting gas

8-month-old girl with 2 days of fever and large-volume loose stools. Sunken eyes, delayed skin turgor, prolonged capillary refill time.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.22	27	11	34	129
K	Cl	Albumin	Glucose	Lactate
3.2	112	3.5	85	1

Plenary reveal - 7 min - Post-intervention gas

Intervention: IV isotonic fluids, correct dehydration.

pH	PaCO ₂	HCO ₃	Na	K
7.35	36	20	135	4.0
Cl	Albumin	Lactate		
105	3.5	1		

Gastroenteritis

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. Primary disorder: HCO_3^- 11 (low) $\rightarrow$ metabolic acidosis.

5. Expected compensation: Winter's: $\text{PaCO}_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$ (22.5-26.5). Actual 27 is at upper edge - borderline compensation.

6. Anion gap: $\text{AG} = 129 - (112 + 11) = 6$ (low/normal). Corrected $\text{AG} = 6 + 2.5 \times (4 - 3.5) = 7.25$ - still normal.

7. Delta ratio: Not applicable - no raised AG.

8. Oxygenation & integration: NAGMA (hyperchloraemic) from stool bicarbonate loss; normal lactate argues against hypoperfusion.

SID / Stewart

SID: $129 - 112 = 17$ - Markedly low - severe chloride-driven acidosis.

Atot: Albumin 3.5 - Mildly low - slightly alkalinising.

Unmeasured anions: $\text{AG} - 1 = 5$ - Normal.

Mechanism: - - Chloride-driven NAGMA.

Post-intervention

6. Anion gap: $\text{AG} = 135 - (105 + 20) = 10$ - normal.

9. SID / Stewart: $\text{SID} = 135 - 105 = 30$ (improving).

Gastroenteritis

Expert comment . answers

Q1. What is the likely cause of the metabolic acidosis in this child?

Q2. How do the chloride and lactate values help you identify the mechanism?

A1. Loss of bicarbonate-rich fluid in large-volume diarrhoea -> hyperchloraemic NAGMA.

A2. Chloride elevated (112) as it replaces lost bicarbonate; normal lactate (1) argues against significant hypoperfusion/shock.

Take-home

A normal-anion-gap acidosis with a normal lactate points to GI bicarbonate loss, not hypoperfusion. SID confirms the chloride-driven mechanism.

Suspected IEM

Small group - 8 min - Presenting gas

8-month-old girl with 1 day of fever and vomiting. Effortless tachypnoea, haemodynamically stable, lethargic. History of similar episodes in the past.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.24	38	16	38	135
K	Cl	Albumin	Glucose	Lactate
4	107	2	85	3

Plenary reveal - 7 min - Post-intervention gas

Intervention: Critical sample, dextrose, stop protein.

pH	PaCO ₂	HCO ₃	Na	K
7.35	35	20	138	4.2
Cl	Albumin	Glucose	Lactate	
105	2.5	100	2	

Suspected IEM

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. Primary disorder: HCO_3^- 16 (low) $\rightarrow$ metabolic acidosis.

5. Expected compensation: Winter's: $\text{PaCO}_2 = 1.5 \times 16 + 8 = 32 \pm 2$ (30-34). Actual 38 - higher than expected $\rightarrow$ additional respiratory acidosis.

6. Anion gap: Raw AG = $135 - (107 + 16) = 12$ (apparently normal). Albumin 2.0: corrected AG = $12 + 2.5 \times (4 - 2) = 17$ (clearly raised).

7. Delta ratio: Using corrected AG: $(17 - 12)/(24 - 16) = 5/8 = 0.625 \rightarrow$ mixed HAGMA + NAGMA.

8. Oxygenation & integration: Concealed HAGMA unmasked only after albumin correction, with inadequate respiratory compensation, in child with recurrent episodes $\rightarrow$ suspect IEM.

SID / Stewart

SID: $135 - 107 = 28$ - Low - chloride-driven component.

Atot: Albumin 2.0 - Markedly low - alkalinising, masking HAGMA.

Unmeasured anions: $17 - 3 = 14$ - Raised.

Mechanism: - - Three forces: low SID, low Atot (masking), unmeasured anions.

Post-intervention

9. SID / Stewart: SID = $138 - 105 = 33$ (normalising). Atot still low (albumin 2.5) - still masking. Unmeasured anions = corrected AG ($20 + 2.5 \times 1.5 = 23.75$) - 2 = 21.75 (still raised).

Suspected IEM

Expert comment . answers

Q1. Calculate the albumin-corrected anion gap.

Q2. Is there an additional respiratory disorder?

Q3. What is the significance of recurrent similar episodes in this clinical setting?

A1. $12 + 2.5 \times (4.0 - 2.0) = 17$ mEq/L. Raw gap of 12 looked normal; low albumin was masking a real HAGMA.

A2. Yes - Winter's predicts PaCO₂ 30-34; actual 38 is higher -> inadequate compensation (superimposed respiratory acidosis).

A3. Recurrent metabolic decompensation triggered by intercurrent illness raises strong suspicion for underlying IEM (organic acidaemia or fatty acid oxidation defect). Prompt metabolic work-up: ammonia, lactate, urine organic acids, plasma amino acids, acylcarnitine - the critical sample.

Take-home

Correct the anion gap for albumin as a reflex, every time - it is most consequential exactly when the raw number looks reassuring. Also check SID.

Salicylate poisoning

Small group - 8 min - Presenting gas

15-year-old boy with tinnitus, fast breathing, vomiting, and confusion after intentional ingestion.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.42	18	11	108	140
K	Cl	Albumin	Glucose	Lactate
4.2	102	4.1	90	3.2

Plenary reveal - 7 min - Post-intervention gas

Intervention: Activated charcoal, IV fluids, sodium bicarbonate, alkalinise urine, consider dialysis.

pH	PaCO ₂	HCO ₃	Na	K
7.48	30	22	142	3.8
Cl	Albumin	Lactate		
105	4.1	1.5		

Salicylate poisoning

Initial . SID . Post-intervention . Q and A

Initial interpretation

- 4. Primary disorder:** Two markedly abnormal primary values (PaCO₂ 18, HCO₃ 11) with near-normal pH - mixed disorder.
- 5. Expected compensation:** If HCO₃ 11 were the only problem, Winter's predicts PaCO₂ 24.5 +/-2. Actual PaCO₂ 18 is far below that - separate primary respiratory alkalosis.
- 6. Anion gap:** AG = 140 - (102 + 11) = 27 (markedly raised). Lactate only partly explains.
- 7. Delta ratio:** (27 - 12)/(24 - 11) = 1.15 -> HAGMA.
- 8. Oxygenation & integration:** Primary respiratory alkalosis + primary HAGMA, opposing pH effects cancelling to near-normal pH - classic salicylate poisoning.

SID / Stewart

- SID:** 140 - 102 = 38 - High-normal - no chloride-driven component.
- Atot:** Albumin 4.1 - Normal.
- Unmeasured anions:** 27 - 3.2 = 23.8 - Markedly raised - salicylate metabolites + ketones.
- Mechanism:** - - Unmeasured anions + primary respiratory alkalosis.

Post-intervention

- 9. SID / Stewart:** SID = 142 - 105 = 37 (normal). Unmeasured anions = AG (142-105-22 = 15) - 1.5 = 13.5 (improving).

Salicylate poisoning

Expert comment . answers

Q1. A near-normal pH can conceal a mixed disorder. What disorders are present here?

Q2. What poisoning is suggested by the clinical history and ABG pattern?

A1. Primary respiratory alkalosis (PaCO_2 far lower than metabolic compensation alone would predict - direct salicylate stimulation of central respiratory centre) + primary HAGMA (AG 27, from lactate, ketosis, salicylate organic acids).

A2. Salicylate (aspirin) poisoning - classic triad of tinnitus, hyperventilation, and mixed respiratory-alkalosis/HAGMA pattern after intentional ingestion in an adolescent.

Take-home

A near-normal pH never means a normal gas - check PaCO_2 and HCO_3^- against each other before you trust it. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.

Post-arrest hanging

Small group - 8 min - Presenting gas

5-year-old boy, 20 kg, accidental hanging. Post-arrest, ROSC after 15 min CPR, ventilated. PEEP 6, PIP 16, RR 20, I:E 1:2, FiO2 100%, TV 80 mL, MAP 17.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
6.88	65	12	70	138
K	Cl	Albumin	Glucose	Lactate
6.2	98	4.1	210	3.2

Plenary reveal - 7 min - Post-intervention gas

Intervention: Increase ventilation, treat hyperkalaemia, fluids, consider bicarbonate.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	40	17	90	138
K	Cl	Albumin	Glucose	Lactate
4.5	100	4.1	180	2.5

Post-arrest hanging

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. **Primary disorder:** Both HCO_3^- 12 and PaCO_2 65 abnormal in acidifying direction - combined metabolic + respiratory acidosis.
5. **Expected compensation:** If pure metabolic, Winter's predicts PaCO_2 26 $\pm$ 2. Actual 65 far above - large separate primary respiratory acidosis.
6. **Anion gap:** $\text{AG} = 138 - (98 + 12) = 28$ (markedly raised). Lactate 3.2 does not fully explain.
7. **Delta ratio:** $(28 - 12)/(24 - 12) = 1.33 \rightarrow$ HAGMA component.
8. **Oxygenation & integration:** $\text{P/F} = 70/1.0 = 70$. $\text{OI} = (17 \times 1.0 \times 100)/70 = 24.3$. Severe ARDS (PALICC $\text{OI} \geq 16$).

SID / Stewart

SID: $138 - 98 = 40$ - High-normal - no chloride-driven component.

Atot: Albumin 4.1 - Normal.

Unmeasured anions: $28 - 3.2 = 24.8$ - Markedly raised - hypoxic-ischaemic anions.

Mechanism: - - Unmeasured anions + primary respiratory acidosis.

Post-intervention

9. **SID / Stewart:** $\text{SID} = 138 - 100 = 38$ (normal). Unmeasured anions = $\text{AG} (138 - 100 - 17 = 21) - 2.5 = 18.5$ (still raised).

Post-arrest hanging

Expert comment . answers

Q1. Identify all components of the mixed acid-base disorder.

Q2. Calculate the P/F ratio and oxygenation index using the available data.

Q3. How would you grade the severity of oxygenation failure?

A1. (1) Primary metabolic acidosis (HCO_3^- 12, AG 28 - hypoxic-ischaemic/lactic) and (2) primary respiratory acidosis (PaCO_2 65, far exceeding expected ~26) - combined metabolic + respiratory acidosis.

A2. $\text{P/F} = 70/1.0 = 70$. $\text{OI} = (17 \times 1.0 \times 100)/70 = 24.3$.

A3. $\text{OI} 24.3$ = severe oxygenation failure (PALICC $\text{OI} \geq 16$ = severe).

Take-home

When PaCO_2 and HCO_3^- are both markedly abnormal in the same direction, name both disorders explicitly - one is not compensating for the other. SID confirms no chloride-driven component.

Septic shock + severe anaemia

Small group - 8 min - Presenting gas

4-year-old boy with septic shock and severe anaemia. FiO2 40%, SpO2 99%, ScvO2 42%.

pH	PaCO2	HCO3	PaO2	Na
7.25	26	11	110	138
K	Cl	Albumin	Hb	Lactate
4.3	106	4.1	4.5	7

Plenary reveal - 7 min - Post-intervention gas

Intervention: Blood transfusion, antibiotics, fluids, vasopressors.

pH	PaCO2	HCO3	PaO2	Na
7.35	36	20	100	138
K	Cl	Albumin	Hb	Lactate
4.0	105	3.8	9.0	2.5
ScvO2				
65%				

Septic shock + severe anaemia

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. Primary disorder: HCO_3^- 11 (low) $\rightarrow$ metabolic acidosis.

5. Expected compensation: Winter's: $\text{PaCO}_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$. Actual 26 - appropriate.

6. Anion gap: $\text{AG} = 138 - (106 + 11) = 21$ (raised). Albumin normal, negligible correction.

7. Delta ratio: $(21 - 12)/(24 - 11) = 0.69 \rightarrow$ mixed HAGMA + NAGMA.

8. Oxygenation & integration: $\text{P/F} = 110/0.4 = 275$. ScvO_2 42% - tissues extracting far more O_2 than usual. $\text{CaO}_2 = (1.34 \times 4.5 \times 0.99) + (0.003 \times 110) = 6.3 \text{ mL/dL}$.

SID / Stewart

SID: $138 - 106 = 32$ - Low - chloride-driven component from resuscitation fluids.

Atot: Albumin 4.1 - Normal.

Unmeasured anions: $21 - 7 = 14$ - Raised - lactic acidosis.

Mechanism: - - Two forces: low SID + unmeasured anions.

Post-intervention

9. SID / Stewart: $\text{SID} = 138 - 105 = 33$ (low-normal). Unmeasured anions = $\text{AG} (138 - 105 - 20 = 13) - 2.5 = 10.5$ (normalising).

Septic shock + severe anaemia

Expert comment . answers

Q1. Calculate the P/F ratio.

Q2. How do you interpret the low ScvO₂ despite a normal SpO₂?

Q3. What do the severe anaemia and elevated lactate indicate about oxygen delivery and extraction?

A1. $110 / 0.4 = 275$ - mild oxygenation impairment; not where the problem lies.

A2. Large veno-arterial oxygen extraction gap: lungs saturating haemoglobin normally, but tissues extracting abnormally high fraction because overall DO₂ inadequate relative to demand. SpO₂ alone misses this.

A3. Severe anaemia critically reduces CaO₂ and therefore DO₂, even though saturation normal. Body compensates initially by increasing extraction (low ScvO₂), but once exceeded, anaerobic metabolism follows - lactate 7. Content/delivery problem, not lung problem. Priority: transfusion.

Take-home

A normal SpO₂ says nothing about oxygen delivery when haemoglobin is this low - look at ScvO₂ and lactate, not the saturation monitor. SID reveals the chloride-driven component.

Methaemoglobinemia

Small group - 8 min - Presenting gas

10-year-old boy with cyanosis, headache, tachycardia, and lethargy after receiving topical benzocaine. Blood sample drawn looks chocolate-brown. FiO2 90%, SpO2 85%, SO2 65%.

pH	PaCO2	HCO3	PaO2	Na
7.36	38	21	550	138
K	Cl	Albumin	MetHb	Lactate
4.3	105	4.1	35%	2

Plenary reveal - 7 min - Post-intervention gas

Intervention: Methylene blue 1-2 mg/kg IV (check G6PD status first), oxygen, stop benzocaine.

pH	PaCO2	HCO3	PaO2	Na
7.38	38	23	200	138
K	Cl	Albumin	MetHb	Lactate
4.0	105	4.1	5%	1.5
SpO2	SO2			
98%	95%			

Methaemoglobinaemia

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. Primary disorder: No significant primary acid-base disorder. Acid-base is distraction.

5. Expected compensation: Not applicable.

6. Anion gap: $AG = 138 - (105 + 21) = 12$ (normal).

7. Delta ratio: Not applicable.

8. Oxygenation & integration: $P/F = 550/0.9 = 611$ (remarkably high). Yet SpO_2 85%, SO_2 65% - large discordant saturation gap. MetHb 35% (normal <1-2%).

SID / Stewart

SID: $138 - 105 = 33$ - Low-normal - mild chloride-driven component.

Atot: Albumin 4.1 - Normal.

Unmeasured anions: $12 - 2 = 10$ - Normal. Acid-base status benign.

Mechanism: - - Acid-base is not the story; oxygenation is.

Post-intervention

9. SID / Stewart: $SID = 138 - 105 = 33$ (low-normal). Unmeasured anions = $AG (138-105-23 = 10) - 1.5 = 8.5$ (normal).

Methaemoglobinaemia

Expert comment . answers

Q1. Calculate the P/F ratio.

Q2. How do you explain the discrepancy between the very high PaO₂, SpO₂, and SO₂?

Q3. What is the significance of the chocolate-brown blood and MetHb 35%?

Q4. What is the most likely diagnosis?

A1. 550 / 0.9 ~611.

A2. PaO₂ measures dissolved oxygen tension and is unaffected by methaemoglobin, so remains high. Methaemoglobin (Fe³⁺, unable to bind oxygen) distorts pulse oximeter algorithm, driving SpO₂ toward falsely fixed value ~82-85%. Directly measured SO₂ by co-oximetry (65%) reflects true, much lower functional saturation - 35% of haemoglobin cannot carry oxygen at all.

A3. Chocolate-brown blood is classic bedside sign. MetHb 35% confirms severe elevation - enough to cause significant functional anaemia (only 65% of haemoglobin can carry oxygen). Explains cyanosis, headache, tachycardia, lethargy despite high PaO₂.

A4. Acquired methaemoglobinaemia (benzocaine-induced).

Take-home

A high PaO₂ does not guarantee oxygen-carrying capacity - whenever SpO₂ and PaO₂ disagree, request co-oximetry before you do anything else.

Two encephalopathic neonates

Small group - 8 min - Presenting gas

Baby A: 4 days old, fed well until day 2, now lethargic and tachypnoeic. Afebrile, clear chest. On antibiotics for sepsis. Baby B: 7 days old, vomiting, lethargic, deep rapid breathing. Platelets 60,000/uL, urine ketones 3+.

A pH	A PaCO ₂	A HCO ₃	A SBE	A Na/Cl
7.51	26	20	-1	138/104
A Glucose	A Lactate	B pH	B PaCO ₂	B HCO ₃
85	1.8	7.14	18	6
B SBE	B Na/Cl	B Glucose	B Lactate	
-21	136/100	45	3.0	

Plenary reveal - 7 min - Post-intervention gas

Intervention: Both: ammonia level, stop protein, dextrose. A: ammonia scavengers if high. B: carnitine, treat acidosis.

A pH	A PaCO ₂	A HCO ₃	A Na/Cl	A Glucose
7.45	35	22	138/105	90
A Lactate	B pH	B PaCO ₂	B HCO ₃	B Na/Cl
1.5	7.30	25	12	138/102
B Glucose	B Lactate			
100	2.0			

Two encephalopathic neonates

Initial . SID . Post-intervention . Q and A

Initial interpretation

- 4. Primary disorder:** A: PaCO₂ 26 (low) -> respiratory alkalosis. B: HCO₃ 6 (low) -> metabolic acidosis.
- 5. Expected compensation:** A: acute resp alkalosis expected HCO₃ ~21.2; actual 20 - appropriate. B: Winter's predicted PaCO₂ 17 +/-2; actual 18 - appropriate.
- 6. Anion gap:** A: AG = 138 - (104 + 20) = 14 (normal). B: AG = 136 - (100 + 6) = 30 (markedly raised).
- 7. Delta ratio:** A: not applicable. B: (30 - 12)/(24 - 6) = 1.0 -> HAGMA.
- 8. Oxygenation & integration:** A: sepsis alone does not explain respiratory alkalosis - suspect hyperammonaemia. B: lactate only 3.0 - rest of gap filled by ketones and organic acids.

SID / Stewart

- A SID:** 138 - 104 = 34 - Low-normal.
- A Unmeasured:** 14 - 1.8 = 12.2 - Normal. Mechanism: respiratory alkalosis.
- B SID:** 136 - 100 = 36 - Normal.
- B Unmeasured:** 30 - 3 = 27 - Markedly raised - organic acids/ketones.

Post-intervention

- 9. SID / Stewart:** A: SID = 138 - 105 = 33; unmeasured anions normal. B: SID = 138 - 102 = 36; unmeasured anions = AG (138-102-12 = 24) - 2 = 22 (still raised).

Two encephalopathic neonates

Expert comment . answers

Q1. Name the primary disorder in each baby. Is compensation appropriate in each?

Q2. Which features does sepsis alone fail to explain?

Q3. Baby B has an anion gap of 30 but lactate is only 3. What fills the rest?

Q4. Intervene: what one test do you send in both, how must the sample be handled, and what do you do while waiting?

A1. A: respiratory alkalosis, appropriate. B: metabolic acidosis, appropriate.

A2. A: respiratory alkalosis without fever or distress. B: severe HAGMA with ketones and hypoglycaemia.

A3. Ketones (urine 3+) and organic acids from inborn error of metabolism.

A4. Ammonia. Sample on ice, delivered to lab within 15 minutes. While waiting: stop protein, give dextrose, prepare ammonia scavengers.

Take-home

Neonatal encephalopathy + abnormal gas = metabolic screen; ammonia on ice is critical. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.

HUS-like

Small group - 8 min - Presenting gas

3-year-old, 5 days after bloody diarrhoea, now oliguric and pale. Hb 6.2 g/dL, platelets 40,000/uL, creatinine 3.8 mg/dL.

pH	PaCO ₂	HCO ₃	SBE	Na
7.22	25	10	-16	131
K	Cl	Albumin	Phosphate	iCa
6.8	104	3.0	9.5	0.92

Plenary reveal - 7 min - Post-intervention gas

Intervention: Dialysis, treat hyperkalaemia and hypocalcaemia.

pH	PaCO ₂	HCO ₃	Na	K
7.35	35	18	135	4.5
Cl	Albumin	Phosphate	iCa	
105	3.2	6.0	1.0	

HUS-like

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. **Primary disorder:** HCO_3^- 10 (low) $\rightarrow$ metabolic acidosis.
5. **Expected compensation:** Winter's: $\text{PaCO}_2 = 1.5 \times 10 + 8 = 23 \pm 2$. Actual 25 - appropriate.
6. **Anion gap:** $\text{AG} = 131 - (104 + 10) = 17$. Albumin-corrected $\text{AG} = 17 + 2.5 \times (4 - 3.0) = 19.5$ (raised).
7. **Delta ratio:** $(17 - 12)/(24 - 10) = 5/14 = 0.36 \rightarrow$ mixed NAGMA + HAGMA.
8. **Oxygenation & integration:** Mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats.

SID / Stewart

SID: $131 - 104 = 27$ - Markedly low - chloride-driven.

Atot: Albumin 3.0 (alkalinising), phosphate 9.5 (markedly acidifying) - Net high Atot.

Unmeasured anions: $19.5 - 0 = 19.5$ - Raised - urate, oxalate, retained organic anions.

Mechanism: - - Three forces: low SID + high Atot (phosphate) + unmeasured anions.

Post-intervention

9. **SID / Stewart:** $\text{SID} = 135 - 105 = 30$ (improving). Atot: phosphate improved from 9.5 to 6.0. Unmeasured anions = $\text{AG} (135 - 105 - 18 = 12) - 0 = 12$ (normalising).

HUS-like

Expert comment . answers

Q1. Is compensation appropriate? What are the anion gap and the albumin-corrected anion gap?

Q2. Calculate the delta ratio. What does it tell you?

Q3. Using SID and Atot, name the three separate forces acidifying this child.

Q4. Intervene: which two sidekick values change your next 10 minutes? The team asks for bicarbonate - what do you say?

A1. Appropriate. AG 17, corrected AG 19.5.

A2. 0.36 -> mixed NAGMA + HAGMA.

A3. Low SID (chloride-driven), high Atot (phosphate), unmeasured anions.

A4. K 6.8 (hyperkalaemia) and iCa 0.92 (hypocalcaemia). Bicarbonate? No - pH 7.22 not severe; bicarbonate can worsen hypocalcaemia and hyperkalaemia.

Take-home

HUS = mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats. SID, Atot, and unmeasured anions each tell a different part of the story.

Hepatitis A encephalopathy

Small group - 8 min - Presenting gas

6-year-old with hepatitis A, grade II encephalopathy. INR 4.2, ammonia 180 umol/L.

pH	PaCO ₂	HCO ₃	SBE	Na
7.49	28	21	-1	132
K	Cl	Albumin	Lactate	Glucose
3.0	98	2.2	5.5	45

Plenary reveal - 7 min - Post-intervention gas

Intervention: Dextrose, lactulose, ammonia scavengers.

pH	PaCO ₂	HCO ₃	Na	K
7.42	32	20	135	3.5
Cl	Albumin	Lactate	Glucose	
100	2.5	3.0	100	

Hepatitis A encephalopathy

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. Primary disorder: PaCO₂ 28 (low) -> respiratory alkalosis.

5. Expected compensation: Acute resp alkalosis: HCO₃ falls ~2 per 10 mmHg fall in PaCO₂. PaCO₂ 28 is 12 below 40 -> expected HCO₃ ~21.6. Actual 21 - appropriate.

6. Anion gap: AG = 132 - (98 + 21) = 13. Corrected AG = 13 + 2.5 x (4 - 2.2) = 17.5 (HAGMA masked by hypoalbuminaemia).

7. Delta ratio: Using corrected AG: (17.5 - 12)/(24 - 21) = 5.5/3 = 1.83 -> mixed.

8. Oxygenation & integration: Respiratory alkalosis + masked HAGMA + hypoglycaemia.

SID / Stewart

SID: 132 - 98 = 34 - Low-normal.

Atot: Albumin 2.2 - Markedly low - alkalinising, masking HAGMA.

Unmeasured anions: 17.5 - 5.5 = 12 - Raised.

Mechanism: - - Masked HAGMA + respiratory alkalosis.

Post-intervention

9. SID / Stewart: SID = 135 - 100 = 35 (normal). Atot still low (albumin 2.5) - still masking. Unmeasured anions = corrected AG (15 + 3.75 = 18.75) - 3 = 15.75 (still raised).

Hepatitis A encephalopathy

Expert comment . answers

Q1. What is the primary disorder, and why is this child alkaline?

Q2. Calculate the anion gap, then correct it for albumin. What changes?

Q3. The base excess is -1, which looks normal. Why is that misleading? How many processes are running at once?

Q4. Intervene: which two sidekicks need treating in the next few minutes, and why does one of them worsen the encephalopathy?

A1. Respiratory alkalosis due to hyperventilation from ammonia stimulating respiratory centre.

A2. AG 13, corrected AG 17.5. Low albumin was masking a HAGMA.

A3. Yes - because HAGMA is masked by hypoalbuminaemia. Multiple processes: respiratory alkalosis, HAGMA (lactic acidosis), hypoglycaemia.

A4. Glucose 45 (hypoglycaemia) and lactate 5.5 (lactic acidosis). Hypoglycaemia worsens encephalopathy.

Take-home

Hepatitis A encephalopathy = respiratory alkalosis + HAGMA; check glucose. Always correct the AG for albumin, and check SID - the SBE may look normal while a HAGMA is masked.

Fasting hypoglycaemia

Small group - 8 min - Presenting gas

18-month-old, previously well, fasted overnight after gastroenteritis and presented with a seizure. Liver palpable 3 cm. ALT 180 U/L, ammonia 110 umol/L, free fatty acids raised.

Glucose	pH	PaCO ₂	HCO ₃	SBE
30	7.36	36	20	-4
Na/Cl	Lactate	b-OHB		
138/105	2.0	0.3		

Plenary reveal - 7 min - Post-intervention gas

Intervention: Critical sample, then dextrose.

Glucose	pH	PaCO ₂	HCO ₃	SBE
100	7.38	38	22	-1
Na/Cl	Lactate	b-OHB		
138/105	1.5	0.2		

Fasting hypoglycaemia

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. **Primary disorder:** HCO_3^- 20 (slightly low), SBE -4 -> mild metabolic acidosis.
5. **Expected compensation:** Not applicable (primary metabolic).
6. **Anion gap:** $\text{AG} = 138 - (105 + 20) = 13$ (normal).
7. **Delta ratio:** Not applicable (NAGMA).
8. **Oxygenation & integration:** Near-normal gas but hypoglycaemia with low ketones -> failure of ketogenesis.

SID / Stewart

SID: $138 - 105 = 33$ - Low-normal - mild chloride-driven component.

Atot: Albumin not given - -

Unmeasured anions: $13 - 2 = 11$ - Normal.

Mechanism: - - Acidosis mild and chloride-driven; the story is ketogenesis failure.

Post-intervention

9. **SID / Stewart:** $\text{SID} = 138 - 105 = 33$ (low-normal). Unmeasured anions = $\text{AG} (138 - 105 - 22 = 11) - 1.5 = 9.5$ (normal).

Fasting hypoglycaemia

Expert comment . answers

Q1. The gas looks almost normal. Why is that the most worrying finding?

Q2. What should a fasting child gas and ketones look like?

Q3. Free fatty acids are high but ketones are low. What does that mean? How would hyperinsulinism differ?

Q4. Intervene: list the critical sample. When exactly must it be drawn, and what do you give immediately afterwards?

A1. In a fasting child, expected ketosis and metabolic acidosis. Normal gas with low b-OHB (0.3) indicates failure of ketogenesis.

A2. Metabolic acidosis with high anion gap due to ketones.

A3. Block in fatty acid oxidation or ketogenesis (e.g., MCAD deficiency). Hyperinsulinism would have low FFA and low ketones.

A4. Must be drawn during hypoglycaemia before treatment. Includes glucose, insulin, C-peptide, cortisol, growth hormone, ketones, free fatty acids, acylcarnitine profile, ammonia, lactate, LFTs. Give dextrose (2-3 mL/kg of 10% dextrose) immediately after.

Take-home

Fasting hypoglycaemia with low ketones = think fatty acid oxidation defect; critical sample before treatment. SID confirms the acidosis was mild and chloride-driven, not the primary problem.

Shocked 2-week-old boy

Small group - 8 min - Presenting gas

2-week-old boy with 3 days of vomiting and poor feeding, now mottled and lethargic. Genitalia look normal. Weight below birth weight.

pH	PaCO ₂	HCO ₃	SBE	Na
7.27	28	12.5	-13	122
K	Cl	Glucose		
7.8	98	40		

Plenary reveal - 7 min - Post-intervention gas

Intervention: Hydrocortisone, fluids with dextrose, treat hyperkalaemia.

pH	PaCO ₂	HCO ₃	SBE	Na
7.35	35	18	-5	135
K	Cl	Glucose		
5.0	100	90		

Shocked 2-week-old boy

Initial . SID . Post-intervention . Q and A

Initial interpretation

4. **Primary disorder:** HCO_3^- 12.5 (low) $\rightarrow$ metabolic acidosis.
5. **Expected compensation:** Winter's: $\text{PaCO}_2 = 1.5 \times 12.5 + 8 = 26.75 \pm 2$. Actual 28 - appropriate.
6. **Anion gap:** $\text{AG} = 122 - (98 + 12.5) = 11.5$ (normal). NAGMA.
7. **Delta ratio:** Not applicable.
8. **Oxygenation & integration:** NAGMA with hyponatraemia, hyperkalaemia, hypoglycaemia $\rightarrow$ CAH salt-wasting crisis.

SID / Stewart

SID: $122 - 98 = 24$ - Markedly low - chloride-driven acidosis.

Atot: Albumin not given - -

Unmeasured anions: $11.5 - 0 = 11.5$ - Normal.

Mechanism: - - Chloride-driven NAGMA.

Post-intervention

9. **SID / Stewart:** $\text{SID} = 135 - 100 = 35$ (normalising). Unmeasured anions = $\text{AG} (135 - 100 - 18 = 17) - 0 = 17$ (raised - resolving).

Shocked 2-week-old boy

Expert comment . answers

Q1. Classify the acid-base disorder and check compensation. Is this a HAGMA or a NAGMA?

Q2. Three sidekick values make the diagnosis before any specialised test. Which, and what is it?

Q3. A vomiting 2-week-old with pyloric stenosis would show which gas instead?

Q4. Intervene: what are your first three actions, and which sample must precede them?

A1. NAGMA with appropriate compensation.

A2. Na 122 (hyponatraemia), K 7.8 (hyperkalaemia), glucose 40 (hypoglycaemia) -> CAH.

A3. Metabolic alkalosis (hypochloraemic, hypokalaemic).

A4. (1) IV access, draw critical sample (electrolytes, glucose, cortisol, ACTH, 17-OHP, renin, aldosterone). (2) Give stress-dose hydrocortisone. (3) IV fluids with dextrose and sodium (normal saline bolus, then D5NS; no potassium initially due to hyperkalaemia). Sample must precede hydrocortisone.

Take-home

Neonatal shock + NAGMA + hyperkalaemia + hyponatraemia = CAH until proven otherwise. SID confirms the chloride-driven mechanism and tracks response to therapy.

Two questions before the post-test

For the whole room

1. Which case changed your mind about a value you thought you understood?
2. Which correction step will you now do as a reflex, every single time - the albumin correction, the SID, or both?